Mhamed Garbaa

AI Engineer – Data Science & MLOps

mhamedgarbaa04@gmail.com Portfolio LinkedIn Tunisia

Professional Profile

Engineer specializing in Artificial Intelligence, Machine Learning, and Data Science. Experience in developing predictive models, analyzing financial data, Explainable AI, and Generative AI. Skilled in Python, PyTorch, scikit-learn, FastAPI, and MLOps, with a focus on experimentation, evaluation, and integration of AI solutions.

Technical Skills

Data Science & ML: Python, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, feature engineering.
Generative AI & NLP: LLMs, Transformers, RAG, LangChain, LangGraph, DPO, LLM evaluation.
Explainable AI: SHAP, LIME, TokenSHAP, counterfactual explanations, LLM-as-a-Judge.
Data & Backend: SQL, PostgreSQL, MySQL, Neo4j, FastAPI, Flask, REST APIs.
MLOps & DevOps: Docker, Git, Linux, CI/CD, MLflow, Azure.

Professional Experience

AI Researcher — LLM Alignment and Evaluation Sep. 2026 – Present

- Contributing to AI research through literature reviews, state-of-the-art analysis, and experimental evaluation of large language models (LLMs).
- Developing reproducible Python pipelines for fine-tuning, preference optimization (DPO), and LLM benchmarking.
- Designing experimental protocols to evaluate model alignment, performance, and reliability, ensuring data traceability and reproducibility of results.

AI Engineering Intern — Agentic AI & Enterprise Systems, Talan Tunisia Feb. – Jun. 2026

- Designed and developed NeoCortex, an intelligent memory platform for AI agents, integrating persistent context and reasoning traceability.
- Architected a hybrid system combining vector search, knowledge graphs (Neo4j), databases (PostgreSQL, LanceDB), and LLM-based reasoning.
- Developed Python/FastAPI services and experimented with retrieval, routing, and contextual reasoning mechanisms for agentic workflows.
- Evaluated context retrieval and utilization mechanisms to improve the relevance and reliability of generated responses.

AI Research Intern — Credit Scoring & Explainable AI, Talan Tunisia Jun. – Aug. 2025

- Designed and developed a Machine Learning-based credit scoring system integrating data preprocessing, risk prediction, and an interactive Gradio interface.
- Studied, implemented, and evaluated various explainability (XAI) methods: SHAP, LIME, TokenSHAP, counterfactual explanations, LLM-as-a-Judge, and mechanistic interpretability approaches.
- Analyzed the fidelity, consistency, and relevance of explanations to assess the transparency and reliability of model decisions.

AI Research Intern — Multimodal Systems, Talan Tunisia Jul. – Aug. 2024

- Integrated processing and interactive visualization components into a multimodal application; prepared technical documentation.

Technical Projects

ECG Analysis — ML Pipeline and Flask Application Academic Project

- Built a Python pipeline for signal preprocessing, feature extraction, and classification using scikit-learn.
- Integrated results into a Flask interface and compared model performance on ECG data.

Driver Distraction Detection — Computer Vision Academic Project

- Developed a real-time detection prototype using Python, OpenCV, MediaPipe, and YOLOv8; tested detection scenarios.

Banking Audit Portal — APIs and Containerization Academic Project

- Designed REST services with access control, logging, and anomaly detection; used Docker for integration.

Education and Certifications

ENET'Com — Telecommunications Engineering Degree, with Honors 2023 – 2026

Coursework in algorithms, machine learning, software development, databases, and systems.
Preparatory Studies: Mathematics and Physics (two years).
Certifications: Microsoft Azure Fundamentals (AZ-900), Azure AI Fundamentals (AI-900), Jenkins CI/CD (KodeKloud), Machine Learning in Production (DeepLearning.AI).

Languages

Arabic (Native) French (B2, TCF-TP) English (C1) German (A2)